

Modeling and Improving Emergency Department Systems using Discrete Event Simulation

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This paper describes a discrete event simulation study of an emergency department at Dr. Georges-L. Dumont Hospital in Moncton (Canada). The objective of the study was to reduce patient waiting times and to improve overall service delivery and system throughput. As patient waiting times are linked to resource availability, a number of alternatives were designed based on adding resource scenarios. Model development was made using Arena software.

Keywords: Emergency department, healthcare modeling, discrete event simulation, decision, waiting time, Arena

1. Introduction

A healthcare system is a pure service system that is characterized by a high human involvement both at resource level (doctor, nurses, etc.) and entity level (patients). Such system can be challenging to design, control or improve when facing the variability resulting from this human involvement. Furthermore, patient diagnosis result depends on several subjective factors and often leads to variability on the amount of resource and on the process route dispatched to patients. This variability leads to longer waiting times and low throughput.

Due to the complexity of healthcare systems, discrete event simulation (DES) has proved to be an effective tool used for process improvement [1, 2], particularly when combined with total quality management (TQM) and continuous quality improvement (CQI) techniques. An emergency department (ED) is the main area where thousands of patients flow every year. Hospital quality of service depends on the performance of its ED. For this reason several studies have been conducted to increase the efficiency of the ED using simulation tools. Most studies found in the literature aim at reducing waiting times and increas-

ing service level (throughput) by improving the actual care process [3, 4] or by increasing the size and the operation of the ED [5, 6].

Komashie et al. [3] conducted a study of an ED in the London area (Britain). The main objective was to determine the impact key resources (doctors, nurses, beds) on key performances (waiting times, waiting queues and throughput). The designed Arena model contains several features such as variable service times to model nurse and doctor ranking along with patient condition-based treatment durations. A different arrival process was used for each day of the week. Data was collected by means of hospital log sheets, interviews, and on-site observations. DES experiments show proportionality between waiting times and doctor utilization. Despite the good quality of the study, several essential elements of EDs were not explicitly included in the model such as lab tests, and triage codes. The highest reduction in waiting time was obtained for a scenario where the system was assumed without blockage so that all patients could be admitted. This scenario is feasible only when a large bed capacity is available. Despite the rare occurrence of this scenario, it remains interesting for simulating health crisis or epidemic situation where system blockage must be avoided.

In an early study Saunders et al. [4] proposed a model that contained several features such as triage priorities, lab tests, teaching aspects, result communication delays, and physicians' collaboration (concurring test results). Patients and lab specimens (blood samples, tests, and re-

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Figures 1, 6–9 appear in color online: <http://sim.sagepub.com>

Table 1. Comparison between previous work and the present study

	Saunders et al. 1989	Samaha et al. 2003	Komashie et al. 2005	Ruohonen et al. 2006	Present study 2007
Arrival process	depends on week days	depends on patient type	depends on week days	depends on patient type	depends on week days
Triage codes	yes, (4 codes)	yes, (4 codes)	no	yes, (4 codes)	yes, (5 codes)
Entities	Patients, lab specimens, test results	Patients	Patients	Patients, lab specimens, test results	Patients
Staff shifts	no	no	yes	yes	yes
Diagnosis or expertise based service times	yes	no	yes	no	yes
Bed ready times	no	yes	no	no	yes
Transfer times	yes	yes	no	no	no
Result transfer times	yes	yes	no	no	yes
Lab tests	yes	yes	no	yes	yes
Teaching and collaboration aspects	yes	yes	no	no	no
Animation	yes	yes	yes	yes	yes
Software	Siman-Cinema	Arena	Arena	Medmodel	Arena

sults) were modeled as entities flowing in the system. The main objective was to study the impact of key resources on waiting times and throughput. It was found that blood test turnaround time has a direct effect on patient throughput. Model logic was implemented using Siman and animation was performed using Cinema.

Ruohonen et al. [5] conducted a study of an ED using Medmodel simulation software (Promodel Corporation) specially designed for evaluation, planning and redesign of hospitals and healthcare systems. The goal was to test the impact of a new triage approach called *triage-team* method on key performances. In the *triage-team* method patient diagnosis is handled by registered nurses and a doctor. Habitually this task was assigned exclusively to registered nurses. Introducing a doctor at the triage level will allow medical diagnosis to be performed and lab tests to be ordered during the early stage of the process. An improvement of waiting times and throughput were observed by simulation. The method allows better priority assignment leading to an accurate and a fast referral to treatment. Resource shifts, lab tests and lab specimen entities were included in the model. Contrary to most studies, arrivals were based on patient types and not on week days.

Samaha et al. [6] analysed the effect of a future ED expansion project on patient stay times. The expansion consists of increasing number of beds, introducing bedside registration and creating a fast-track centre. Two states were characterized, namely, bypass state and divert state. The former corresponds to a saturated ambulatory service and the later to a saturated room service. No admission was possible in either case. The goal was to achieve an expansion that will prevent the occurrence of these states. The simulation model included features such as bed ready

times, transportation delays, and delays associated with teaching aspect that requires medical students' presence during patient treatment. It was not mentioned if service times were varied to consider ranking between novice and expert physicians. The model incorporated lab tests and triage codes. The alternative with fast-track centre allows the achievement of better performance by expediting non-critical patients through the system and shortening their length of stay in ED. Adding beds and bedside registration turns out to be not profitable for the level of improvement achieved. Only patients arriving by car, ambulance or from trauma were included in the study. It was not mentioned if the arrival process was based on patient types or on days of the week.

This paper studied an ED of a regional hospital to improve its current operation. Our methodology was based on considering key resources (physicians, nurses, and examination rooms) as control variables. The approach was to use data collection and simulation to build a valid model of the process. The simulation model was implemented using Arena software (Rockwell Softwares Corporation). Table 1 shows all features included in our model compared to previous studies.

The paper is organized as follow: an overview of the ED is given in Section 2. The methodology adopted for modeling and simulation is explained in Section 3. Data collection approach and input analysis is given in Section 4. Section 5 presents the modeling approach and output analysis for model validation. Section 6, presents a set of alternatives designed to improve waiting time using staff and examination rooms as control variables. Results are presented in Section 7 and the best alternative is selected. A number of qualitative improvement suggestions are presented in Section 8, and conclusion in Section 9.

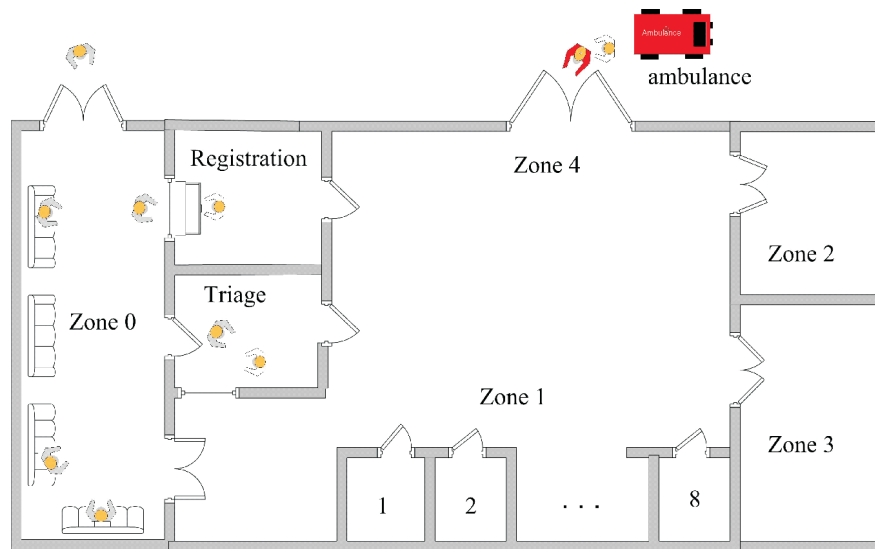


Figure 1. Emergency department overview

2. Emergency Department Overview

The Dr. Georges-L. Dumont Regional Hospital is the main component of the Beauséjour Regional Health Authority. This 302-bed hospital provides primary, specialized and tertiary (more complex) health care services, with the support of leading edge technology. This hospital is a provincial referral centre for its specialized services. It employs more than 130 doctors, 1700 employees and 300 volunteers. Its mission is to become a centre for academic excellence in health sciences and health care services for all of Atlantic Canada. The ED is responsible for triage and for treatment of patients arriving at the Dr. Georges-L. Dumont Regional Hospital. Open 24 h a day, the ED has more than 50 000 patient visits per year. The department has a 16-bed capacity; eight of these beds are reserved for accident victims and critical care and seven are reserved for patients who are acutely ill. The ED, along with several multidisciplinary teams, participates in community projects such as suicide prevention, and crisis intervention.

2.1 Physical Layout

The physical layout is shown in Figure 1. There are five distinct zones in the ED: mobile patient arrival and waiting area (zone 0), examination rooms (zone 1), inpatient care unit intensive care unit (ICU) room (zone 2), observation rooms (zone 3), and ambulance arrival area (zone 4). Critical patients arriving by ambulance are admitted first to the ICU room (trauma). Examination rooms are used to treat patients arriving from zone 0, while observation rooms are for ill patients who require their condition to be monitored. There are eight general examination

rooms, four are equipped to deliver specific care such as orthopaedics, hand surgery, infections treatment, and ophthalmology.

2.2 Healthcare Staff and Shifts

The 24-h operation of the ED requires one nurse at triage and one at registration. Their shift is 8 h starting at midnight, 0800 h and 1600 h. There are five physicians: three with an 8-h shift starting at midnight, 0800 h and 1600, and two with 7 and 6-h shifts starting respectively at 1000 h and 1700 h. Overlapping shifts make it possible to have more than one physician at specific times. Examination room nurses work in conjunction with physicians and share similar shifts.

2.3 Patient Flow

The random flow of patients has three main characteristics. First, it is driven by seasonal illness or incident. In winter most patients are admitted for cold, lung infection illness, and flu viruses, whereas in summer admissions are often related to outdoor incidents and allergies. Second, the flow fluctuates considerably depending on days of the week: more arrivals are observed on Mondays and Fridays than the rest of the week (this may be due to fatigue appearing usually in the end of the week). Third, patient arrivals increase between 0800 and 2000 h of a given day. Figure 2 confirms this observation as it shows patient arrival pattern based on data collected from a study conducted on 163 Canadian Hospitals [7].

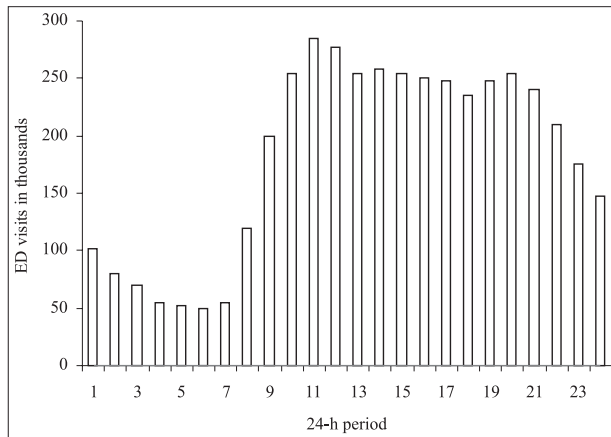


Figure 2. Patient arrival pattern over a 24-h period (source: [7])

2.4 Process Stages

The process stages are shown in Figure 3. A regular patient enters the system at zone 0, picks a number and remains in the waiting area. When his number is called the patient is assessed by a triage nurse who screens for critical apparent symptoms (high blood pressure, fever, etc.). If patient is found to be in critical condition, he is transferred to the ICU room for immediate care. Otherwise, the triage nurse assigns a triage code depending on the patient condition (1 to 5, 1 being most critical).

Once a triage code is assigned, the patient waits for physician assessment. The waiting period depends on the availability of physicians and examination rooms. After his first assessment, lab tests may be required by the physician (blood test, lung scanner, etc.). If not, the patient is discharged to go home or transferred to another department for admission. A patient with lab test results has to wait again for a second assessment by the physician requiring the tests. After the second assessment, the patient may be discharged to go home with a prescription or transferred for admission. Patients arriving by ambulance are transferred directly to the trauma room without triage.

2.5 Triage Codes

Triage codes are critical to achieve a correct dispatch of patients (giving the right priority based on patient condition). In Canada, waiting times associated with each code are standardized. Patients assigned code 1 undergo zero waiting time (critical condition); patients with code 2, 3, 4 and 5 should not wait more than 15, 30, 60, and 120 min respectively [7]. We collected waiting times by code over 90 days, between 0800 and 2000 h. Mean values (in min.) are presented in Table 2.

Table 2. Average waiting times by code

	Waiting times by week days					Canadian standards
	Mon	Tue	Wed	Thu	Fri	
Code 1	0	0	0	0	0	0
Code 2	0	0	4	0	0	15
Code 3	74	47	52	30	71	30
Code 4	213	128	157	100	119	60
Code 5	283	160	184	117	136	120

Patient flow is dispatched by the triage service according to the following rates (collected during the same period): 18% code 3, 54% code 4, 20% code 5, and less than 7% for codes 1 and 2. Table 2 shows that waiting time standards were not met for code 3, 4 and 5. These delays affect about 93% of patients visiting the ED.

3. Proposed Methodology

In this study our goal was to improve the care process by bringing waiting times to their standard values or below. Our methodology was based on considering physicians, nurses, and examination rooms as control variables. This approach is similar to previous studies proposed in [3] and [4]. Several alternatives were designed for different variable settings. DES was then used to evaluate key performance improvements. This can be justified since patient waiting time depends directly on staff and examination room availability. The simulation stage was based on a valid process model. The main steps of the proposed methodology are shown in Figure 4.

Choice and size of data to collect is critical for model design and validation. The data are time durations collected at different process steps through on-site observations and interviews. This is explained in the following section.

4. Data Collection and Input Analysis

Collected time durations were of two types: waiting durations and activity durations. Waiting duration is the time spent by a patient from the end of an activity or an event to the start of the next activity in the process. Activity duration is the time required to complete an activity in the process. Table 3 shows the different durations collected.

Other activity durations can be deduced from the above times: triage time ($T_2 - T_1$), and registration time ($T_3 - T_2$). Duration T_3 depends on codes assigned by triage. Patient arrival times were also collected for different week days. For each data, a sample size of 640 was collected over 90 days, between 0800 and 2000 h. The period of the day was suggested by the quality management team as it is the busiest according to Figure 2. Using Arena software's

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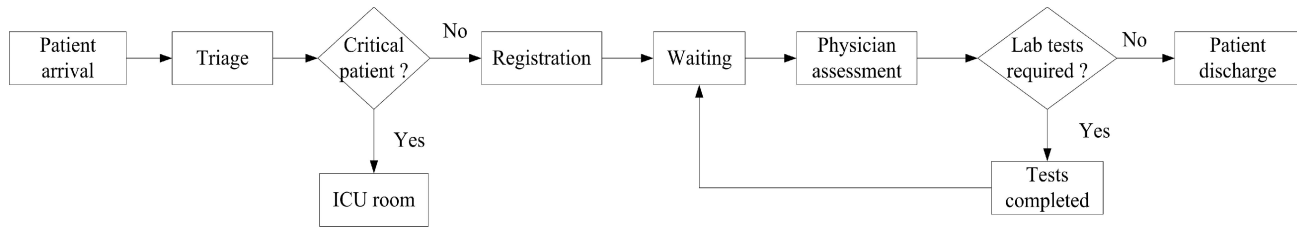


Figure 3. Care process in the emergency department

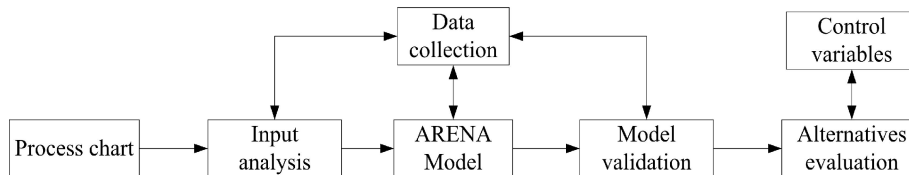


Figure 4. Methodology main steps

Table 3. Collected waiting and activity durations

Waiting durations		Activity durations	
T1	Time between arrival and triage	T6	Time required for lab tests completion
T2	Time between triage and registration	T7	Time required for physician assessment
T3	Time from registration to available exam room	T8	Exam room ready-time
T4	Time from first assessment to discharge		

Table 4. Fitting statistical distributions (time unit is minutes)

Patient arrival			Physician assessment		Process activities	
			First visit	Second visit		
Mon	EXPO (7.0)	code3	TRIA(25,30,40)	TRIA(10,12,15)	Triage	POIS(6)
Tue	EXPO (9.5)	code4	TRIA(25,30,40)	TRIA(8,10,12)	Regist.	TRIA(3,5,7)
Wed, Thu, Fri	EXPO (10)	code5	TRIA(25,30,40)	TRIA(6,7.5,9)	Lab tests	TRIA(30,45,60)

input analyzer, we identified fitting distributions (all P-values > 0.1) for each data set (Table 4).

Based on the collected observations, 77% of patients were treated by one visit (no lab test needed) and 23% needed lab tests, so they required a second assessment by the same physician ordering the tests. TRIA, POIS and EXPO are abbreviations for Triangular, Poisson and Exponential distributions respectively. All time parameters are in minutes.

5. Model Design and Validation

A number of hypotheses were made to define model limits based on the available data. These hypotheses are listed here.

- Only weekdays (Monday to Friday) were considered.
- For a given day the process was simulated only from 0800 to 2000 h.
- Time delays were supposed to depend only on nurses, physicians or rooms. Other sources were not considered (janitorial staff, lack of information on a patient, computer system, maintenance, etc.).
- Patient waiting time from entering room to physician arrival was not considered (difficult to collect).
- Nurses and physicians work in conjunction: when a physician is assigned or released, a nurse is also assigned or released. If a new physician is added to the staff, a new nurse is also added.

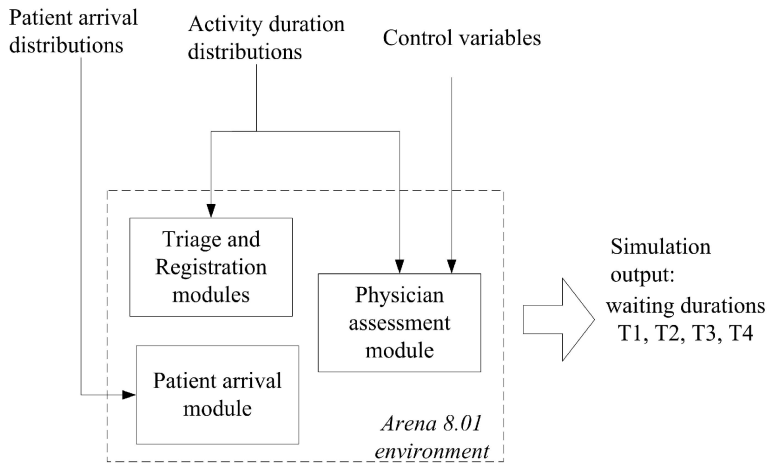


Figure 5. Emergency department's Arena model

Table 5. Simulation output vs. real collected data (time in minutes)

Days	Duration T1			Duration T2			Duration T3			Duration T4		
	M3	S3	R3	M3	S3	R3	M3	S3	R3	M4	S4	R4
Mon	13.6	5.7	12.7	1.4	0.5	1.7	200.9	15.4	235.0	58.2	6.8	36.0
Tue	4.7	1.2	6.6	0.8	0.2	0.6	139.8	18.4	144.0	49.9	4.7	36.0
Wed	4.2	1.3	10.0	0.7	0.2	1.8	115.5	32.6	121.0	56.7	5.8	40.0
Thu	4.2	1.3	10.0	0.7	0.2	1.8	115.5	32.6	121.0	56.7	5.8	40.0
Fri	4.3	1.3	17.9	0.5	0.2	2.2	111.2	34.9	101.0	60.6	10.3	42.0

- (f) Only five of the eight examination rooms were considered as the other rooms are reserved for specific situations.

The limitations of the model were due to the following reasons.

- (a) Data were collected during summer time. Model output can not be generalized to other seasons. Thus, the seasonality dynamic could not be modeled.
- (b) Flow of patient assigned triage code 1 or code 2 was not considered. This flow represented less than 7% of the total flow.
- (c) All transfer times (travel times) in the ED were neglected.
- (d) During Saturday and Sunday, arrivals were very slow compared to other days of the week. They were omitted from the model because the ED never reached its steady state during these days. This will avoid type I & II errors: small time values during weekend would average out larger values occurring during week days.

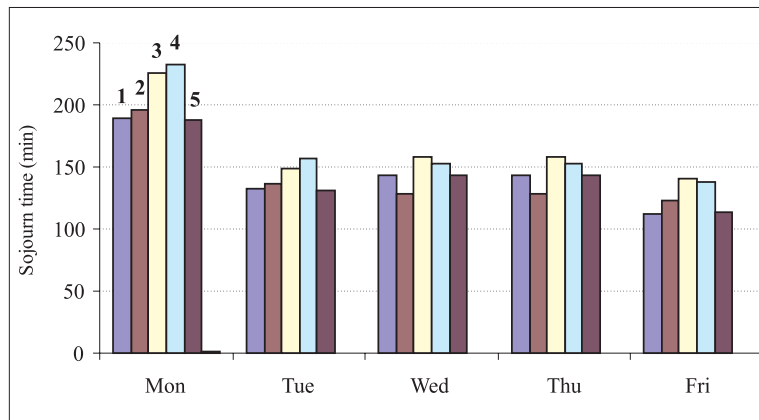
A modular approach was adopted during model development. The final model is made of three modules: patient arrival module, triage and registration module, and physician assessment module. The later contains lab tests and patient discharge sub-modules. Model output consists of waiting durations T1, T2, T3 and T4 (Figure 5).

The model was simulated for every week day. Ten replications of 12 h were performed for each day. The system was always initialized between replications. Simulation output was compared with the collected data for model validation, using a confidence level of 95%. The comparison is summarized in Table 5. Mean value and standard deviation of simulated T_i ($i = 1, 2, 3, 4$) are designated by M_i , and S_i , respectively. Real collected mean value is designated by R_i .

The largest divergence from real data was observed on T3. For instance, $R3 - M3 = 35$ min for Monday, corresponds to a relative error of 15%. Increasing the number of replications shows no significant effect on this error. Thus, it is more likely to be caused by off-sets in the model (systematic errors). This can be improved by reconsidering patient transfer times, or other assumptions and limitations made earlier. The model remains practical for decision making. It provides different outputs such as queue times, queue sizes, and resource utilizations. These out-

Table 6. Alternatives and their corresponding resource increases

	Control variables		
	Additional nurse, shift	Additional physician, shift	Additional room
Alternative 1	1 nurse, [0800 h, 1600 h]	1 physician, [0800 h, 1600 h]	0
Alternative 2	1 nurse, [0800 h, 1600 h]	1 physician, [0800 h, 1600 h]	1
Alternative 3	1 nurse, [1000 h, 1700 h]	1 physician, [1000 h, 1700 h]	0
Alternative 4	1 nurse, [1000 h, 1700 h]	1 physician, [1000 h, 1700 h]	1
Alternative 5	1 nurse, [0800 h, 1600 h] 1 nurse, [1600 h, 2000 h]	1 physician, [0800 h, 1600 h] 1 physician, [1600 h, 2000 h]	0

**Figure 6.** Sojourn time during the week for alternatives 1, 2, 3, 4 and 5

puts were also presented to hospital staff along with the animation for validation purpose. Our prime focus was to target waiting duration T3, as it constitutes the largest portion of the total waiting time in the process. In the next section, five alternatives are identified to improve T3 using the simulation model.

6. Design of Alternatives

The design of the alternatives is based on considering physicians, nurses, and examination rooms as control variables. Initially, the ED employs five physicians, five nurses, three triage nurses and three registration nurses starting all at their respective shift. It also contains eight examination rooms but operates only with five. The quality management team at the hospital was interested in finding an economical way to reduce waiting time by increasing staff and/or room capacity within budgetary constraints. We designed the following five alternatives accordingly (Table 6).

Each alternative was simulated for each day using ten replications of 12-h lengths. Patient sojourn time in the ED (time spent in the system from entrance to discharge) was collected using simulation tally. This time was equivalent to the sum of T_i ($i = 1, 2, 3, 4$). Room usage and

number of patients treated in a day were also collected. The results are summarized and discussed in the next section.

7. Simulation Results and Discussion

Figures 6 and 7 show respectively patient sojourn times and waiting times T3. As we mentioned earlier, time T3 constitutes the largest portion of the total waiting time in the ED. Larger T3 values result in a longer sojourn time and substandard quality service.

Alternatives 1, 2 and 5 were the ones that yielded a significant improvement of sojourn time and waiting duration (Figure 7). A decrease of up to 2 h was achieved by these alternatives compared to the actual situation in the ED. Alternative 1 is more interesting economically since it requires only two medical staff (one physician and one nurse) and no exam room. In contrast, alternative 5 was the least attractive financially as it requires four additional staff.

One consequence of shorter waiting time is the increase of service level (number of patients treated) (Figure 8). Alternatives 1 and 2 have similar service level. Thus, adding an exam room without a corresponding increase in medical staff will have close to no effect on the service level

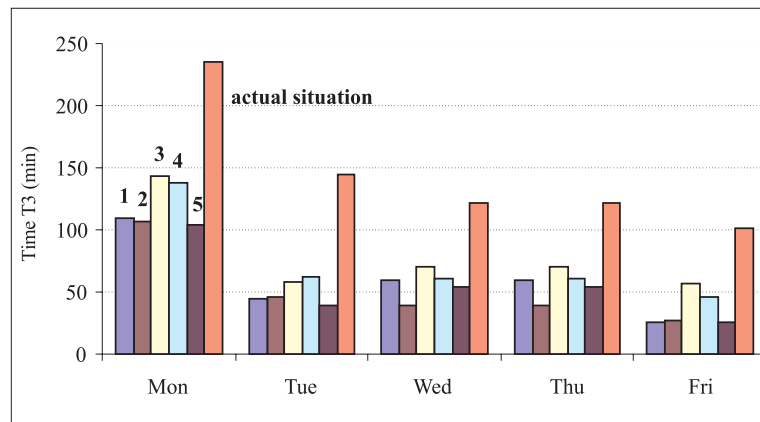


Figure 7. Waiting time T3 per alternative vs. actual value

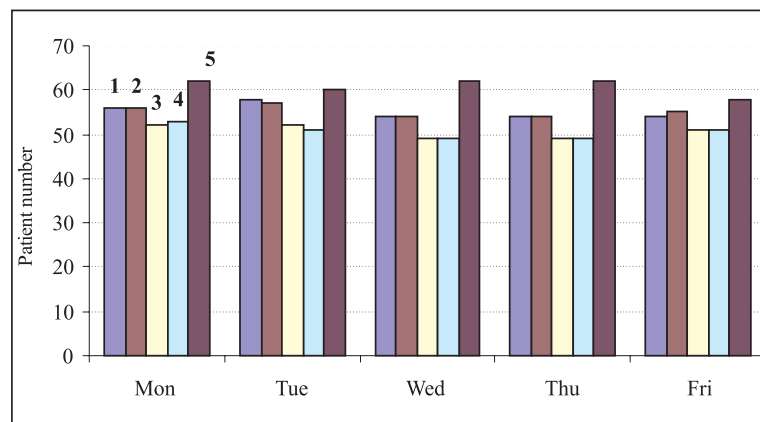


Figure 8. Patient number treated during week days for each alternative

improvement. Actually, alternative 5 shows, that more patients can be treated with additional staff. Even if, alternative 5 exceeds in service level it remains a slight improvement (five to eight more patients) when compared with alternative 1. The difference in service level is small for justifying alternative 5 economically.

Exam room usage rates are represented by Figure 9. Alternatives 1 and 2 give comparable waiting time and service level. Adding a room does not give a significant improvement on these two performances. According to Figure 9, from alternative 1 to 2, adding a room shows a decrease in room usage. Similar remarks apply for alternatives 3 and 4. Adding exam rooms without matching increases in staff makes no significant improvement to waiting time. This result complies with the findings of [3], [4] and [6]. An additional room also implies cleaning and maintenance costs that make alternatives 2 and 4 less interesting compared, respectively, to alternatives 1 and 3.

Considering the budget limit, the ED management team opted for alternative 1. This alternative allows the

best improvement of sojourn time and global waiting time: waiting time is reduced up to 2 h at the beginning of the week. Furthermore, it increases the number of patients treated between 0800 and 2000 h by an average of 16 patients. This has a positive impact on the service level of the ED. It also shows that significant waiting time is not caused by exam room shortage as suspected initially by the management team. Actually, alternative 1 uses only five out of the eight rooms available, and the usage rate is still at 90%.

8. Continuous Quality Improvement

The simulation project conducted for the ED of Dr. Georges-L. Dumont Hospital allowed us to capture and learn more information about the process. DES is not only about data collection and output analysis but it involves learning the complexity of the system and designing a valid model that is useful for training and decision mak-

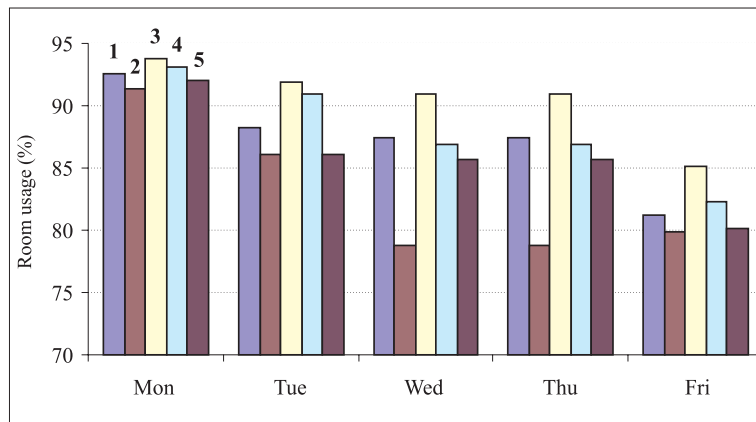


Figure 9. Exam rooms usage during the week for each alternative

ing. This experiment helps us to suggest the following qualitative improvements.

- (a) The use of a ticket dispenser with a built-in time clock, will allow an accurate estimation of T1 since it prints the arrival time on pulled out tickets. This time can be considered by the triage nurse when assigning codes.
- (b) An additional nurse can be hired to assist patients fill up information sheets regarding medical condition and drug history. The sheet can be presented at the triage or registration to speed up the process.
- (c) Make a suggestion poll available to allow every member of staff and patient to participate in the general improvement process of the ED.
- (d) Use survey before discharging patients to collect difficulties experienced throughout the process.
- (e) Computerize patient files to allow fast processing and communication between different services of the ED
- (f) An additional standby triage station can allow a better priority dispatching when patient flow is important (day's afternoon and beginning of the week).

9. Conclusion

In this study an ED of a regional hospital was modeled, analysed and improved using discrete simulation. An analysis of waiting times by patient codes demonstrated high waiting times in comparison with Canadian standards. Thus, the goal of the study was to reduce waiting times by considering what-if analysis using simulation. Only codes 3 to 5 were considered for two reasons.

First, this constitutes 93% of patients visiting the emergency. Second, code 1 and 2 were conforming to standards. Several data were collected from Monday to Friday between 0800 and 2000 h. Weekend was not considered for its slow dynamic to avoid type I & II errors. Collection period choice was based on historical data and staff expertise. Modeling and simulation were performed using Arena software. Simulation shows that waiting duration from registration to available exam room (T3) was the most problematical (3 to 4 h at times). Five alternatives were formulated based on adding staff and exam rooms within budget limitation. The alternative with one physician and nurse added from 0800 to 1600 h gave the best improvement level for waiting time T3. It allows treating an additional number of 16 patients between 0800 and 2000 h. Simulation shows that number of examination rooms had no effect on waiting time if added without a matching increase in the staff. A number of qualitative suggestions were also formulated based on our experience with the process. Even if these suggestions concern the studied ED, they still can be useful for many other health-care systems facing similar problems.

10. Acknowledgement

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